Beyond the Spreadsheet

Modernizing Conjoint Simulators Using a Shiny Toolkit

KS&R

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Quant UX Con 2026

Conjoint turns trade-offs into decision models

Choice-based conjoint (CBC) recreates realistic trade-off decisions to reveal the preferences underneath.

A realistic choice task

Which ice cream sundae would you choose?

SUNDAE 1	SUNDAE 2	SUNDAE 3
FLAVOR	FLAVOR	FLAVOR
Strawberry	**Vanilla**	**Chocolate**
TOPPINGS	TOPPINGS	TOPPINGS
Caramel sauce	Hot fudge + Sprinkles	Sprinkles
SIZE	SIZE	SIZE
3 scoops	**1 scoop**	**2 scoops**
PRICE	PRICE	PRICE
$5.99	**$4.99**	**$4.99**
○ Select	● **Selected**	○ Select

○ None of these sundaes

HOW CBC WORKS

1. **Show alternatives**
   Different combinations of features and levels.
2. **Ask respondents**
   Choose one sundae (or none).
3. **Repeat choices**
   Learn which trade-offs matter.
4. **Model preferences**
   Estimate the value of each level.

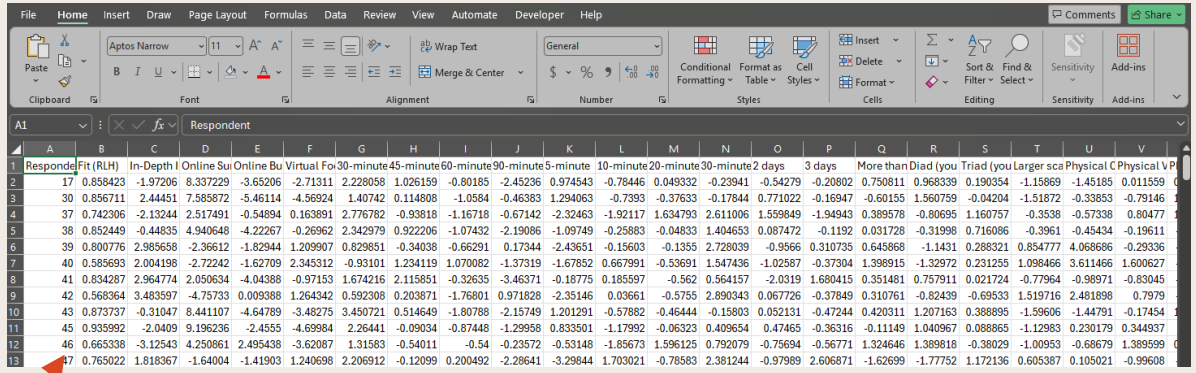
THE RESULT **A model of how customers make choices and tradeoffs, not just what they say is important.**

The conjoint is the model.

The model estimates trade-offs.

WHAT CONJOINT DOES

- **Uses experimentally designed choice activity**
  Asks respondents to make choices among various alternatives
- **Quantifies trade-offs**
  Reflects what people give up to get something else.
- **Captures substitution and cannibalization**
  Accounts for choices among competing alternatives.
- **Estimates preferences**
  Quantifies the relative appeal of tested levels via utility estimates derived from modeling the choice data



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	Responde	Fit (RLH)	In-Depth I	Online Su	Online Bu	Virtual Fo	30-minute	45-minute	60-minute	90-minute	5-minute	10-minute	20-minute	30-minute	2 days	3 days	More than	Diad (you	Triad (you	Larger sca	Physical C	Physical V
2	17	0.858423	-1.97206	8.337229	-3.65206	-2.71311	2.228058	1.026159	-0.80185	-2.45236	0.974543	-0.78446	0.049332	-0.23941	-0.54279	-0.20802	0.750811	0.968339	0.190354	-1.15869	-1.45185	0.011559
3	30	0.856711	2.44451	7.585872	-5.46114	-4.56924	1.40742	0.114808	-1.0584	-0.46383	1.294063	-0.7393	-0.37633	-0.17844	0.771022	-0.16947	-0.60155	1.560759	-0.04204	-1.51872	-0.33853	-0.79146
4	37	0.742306	-2.13244	2.517491	-0.54894	0.163891	2.776782	-0.93818	-1.16718	-0.67142	-2.32463	-1.92117	1.634793	2.611006	1.559849	-1.94943	0.389578	-0.80695	1.160757	-0.3538	-0.57338	0.80477
5	38	0.852449	-0.44835	4.940648	-4.22267	-0.26962	2.342979	0.922206	-1.07432	-2.19086	-1.09749	-0.25883	-0.04833	1.404653	0.087472	-0.1192	0.031728	-0.31998	0.716086	-0.3961	-0.45434	-0.19611
6	39	0.800776	2.985658	-2.36612	-1.82944	1.209907	0.829851	-0.34038	-0.66291	0.17344	-2.43651	-0.15603	-0.1355	2.728039	-0.9566	0.310735	0.645868	-1.1431	0.288321	0.854777	4.068686	-0.29336
7	40	0.585693	2.004198	-2.72242	-1.62709	2.345312	-0.93101	1.234119	1.070082	-1.37319	-1.67852	0.667991	-0.53691	1.547436	-1.02587	-0.37304	1.398915	-1.32972	0.231255	1.098466	3.611466	1.600627
8	41	0.834287	2.964774	2.050634	-4.04388	-0.97153	1.674216	2.115851	-0.32635	-3.46371	-0.18775	0.185597	-0.562	0.564157	-2.0319	1.680415	0.351481	0.757911	0.021724	-0.77964	-0.98971	-0.83045
9	42	0.568364	3.483597	-4.75733	0.009388	1.264342	0.592308	0.203871	-1.76801	0.971828	-2.35146	0.03661	-0.5755	2.890343	0.067726	-0.37849	0.310761	-0.82439	-0.69533	1.519716	2.481898	0.7979
10	43	0.873737	-0.31047	8.441107	-4.64789	-3.48275	3.450721	0.514649	-1.80788	-2.15749	1.201291	-0.57882	-0.46444	-0.15803	0.052131	-0.47244	0.420311	1.207163	0.388895	-1.59606	-1.44791	-0.17454
11	45	0.935992	-2.0409	9.196236	-2.4555	-4.69984	2.26441	-0.09034	-0.87448	-1.29958	0.833501	-1.17992	-0.06323	0.409654	0.47465	-0.36316	-0.11149	1.040967	0.088865	-1.12983	0.230179	0.344937
12	46	0.665338	-3.12543	4.250861	2.495438	-3.62087	1.31583	-0.54011	-0.54	-0.23572	-0.53148	-1.85673	1.596125	0.792079	-0.75694	-0.56771	1.324646	1.389818	-0.38029	-1.00953	-0.68679	1.389599
13	47	0.765022	1.818367	-1.64004	-1.41903	1.240698	2.206912	-0.12099	0.200492	-2.28641	-3.29844	1.703021	-0.78583	2.381244	-0.97989	2.606871	-1.62699	-1.77752	1.172136	0.605387	0.105021	-0.99608

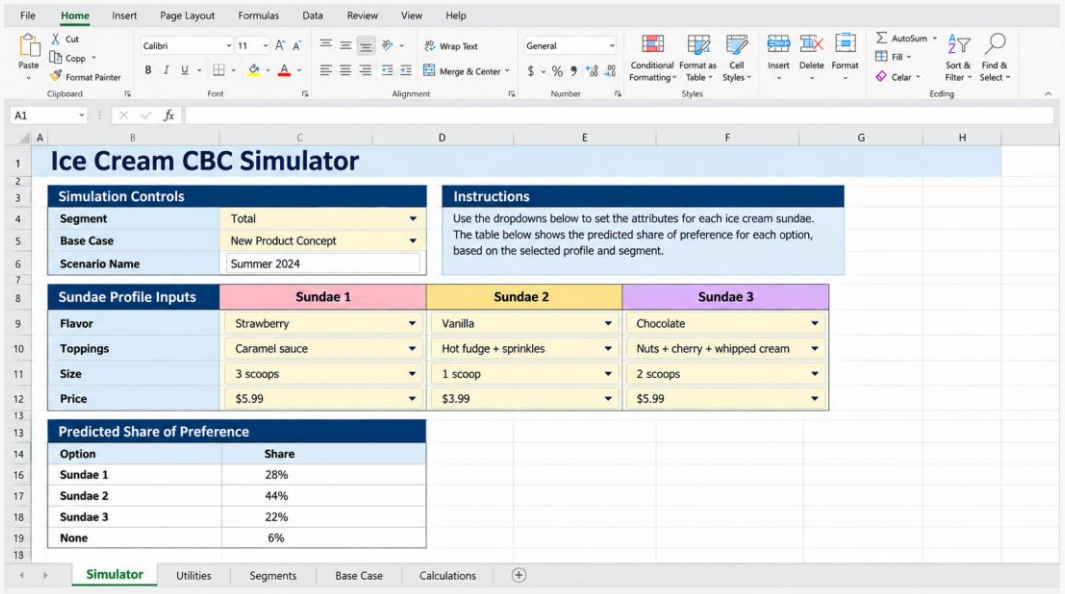
The Model

- HB / respondent-level utilities estimated from choice activity
- Conjoint also allows for downstream analyses to support deeper insights
  - Attribute importance
  - Calibrated outputs
  - Price elasticity
  - Segment or subgroup analysis

The simulator is the deliverable.

The simulator turns the estimates into something stakeholders can explore.

A familiar deliverable: the Excel simulator



Ice Cream CBC Simulator

Simulation Controls	
Segment	Total
Base Case	New Product Concept
Scenario Name	Summer 2024

Instructions

Use the dropdowns below to set the attributes for each ice cream sundae. The table below shows the predicted share of preference for each option, based on the selected profile and segment.

Sundae Profile Inputs	Sundae 1	Sundae 2	Sundae 3
Flavor	Strawberry	Vanilla	Chocolate
Toppings	Caramel sauce	Hot fudge + sprinkles	Nuts + cherry + whipped cream
Size	3 scoops	1 scoop	2 scoops
Price	$5.99	$3.99	$5.99

Predicted Share of Preference	
Option	Share
Sundae 1	28%
Sundae 2	44%
Sundae 3	22%
None	6%

Simulator | Utilities | Segments | Base Case | Calculations

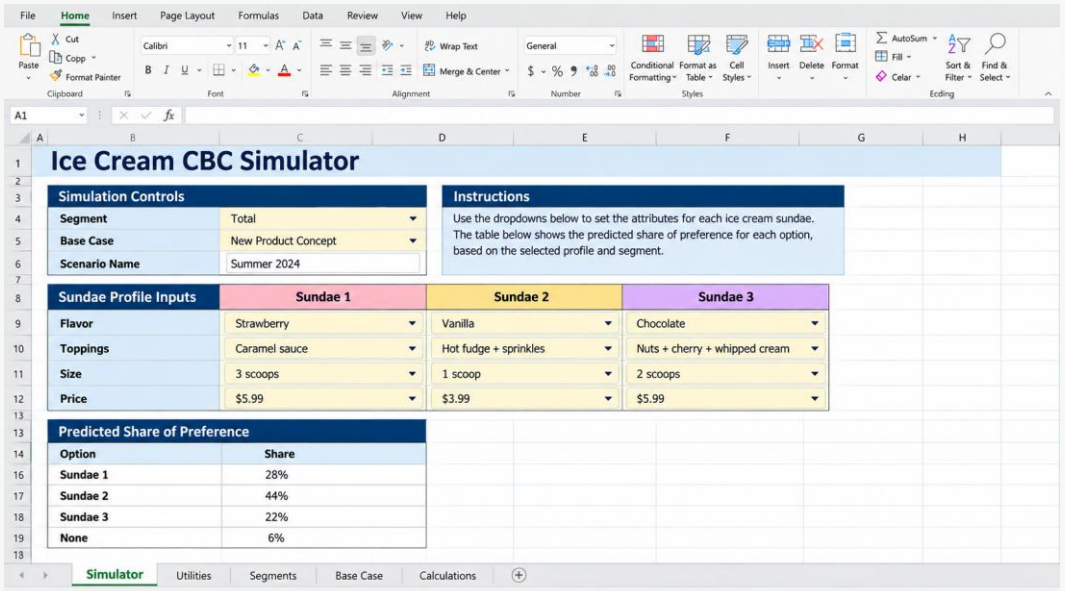
WHAT THE SIMULATOR DOES

- **Applies the model**
  Uses estimated utilities to calculate share of preference (SOP)
- **Configures scenarios**
  Create various combinations of tested attribute levels, prices, or conditions.
- **Explores what changes**
  Compare baselines, segments and alternatives.

The Deliverable Gap

The conjoint method has evolved. Stakeholder-facing simulators have lagged behind

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Simulator | Utilities | Segments | Base Case | Calculations

Functional? Yes, Stakeholder-first? Not always

Less Intuitive for broader stakeholder use
Inputs, assumptions, and design constraints are not always easy to infer

Guardrails
Cumbersome to implement guardrails preventing invalid scenario combinations or price levels outside of design scope

Maintenance
Version control, formula protection, and repeated custom outputs can create overhead.

Modernizing the simulator benefits researchers

Leveraging a Shiny infrastructure to create the simulator can make research teams more effective

FOR THE RESEARCHER

Automate analytical workflows

Connect scenario inputs, validation, and calculations in a single standardized workflow. Export scenario configurations and results to streamline reporting and comparisons.

Centralize additional analyses & outputs

Integrate segment comparisons, attribute importance, calibration, and other analyses directly into the simulator rather than creating separate deliverables.

Simplify updates and maintenance

Make fixes or adjust scenario rules in one application rather than maintaining and redistributing multiple spreadsheet versions.

Benefits clients and other stakeholders too!

The same simulator application also makes the research easier to explore and act on.

FOR THE CLIENT / STAKEHOLDER

Intuitive, user-friendly interface

Explore relevant scenarios with clear controls, meaningful comparisons, and helpful feedback.

Stay within guardrails

Get clear guidance when scenarios exceed the model's supported ranges or violate feasibility rules.

Broader stakeholder access

Enables stakeholders across teams to independently explore conjoint results, test scenarios, and answer questions via a guided, self-service experience.

Move from scenarios to decisions

Export scenario configurations and results to easily compare key alternatives and share findings.

Applying UI Principles in an Example Shiny Simulator

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Practical UI principles for a Shiny conjoint simulator

A purpose-built Shiny app makes it easier to encode thoughtful interaction patterns into the deliverable.

01 **Show relevant inputs**

Progressive disclosure (only show what matters now)

Relevant, context-aware controls & messaging

02 **Make the structure obvious**

Visual hierarchy & grouping

Distinct input, comparison and output areas with clear labels.

03 **Provide helpful feedback**

Validation & recovery helps users avoid mistakes

Bounds, logical restrictions and informative error messages.

04 **Encourage exploration**

Easy user control and reversibility encourage exploration

Sensible defaults, flexible scenario adjustments, and allow an easy return to base case

Demo: UI principles applied in a simulator

Show relevant inputs | **Make structure obvious** | **Provide helpful feedback** | **Encourage exploration**

WORKBOOK-STYLE EXPERIENCE

Flavor	Strawberry	Holiday Special
Price	$5.99	$5.79
Number of Scoops	3 Scoops	1 Scoop
Toppings	Sprinkles	Caramel

MODERNIZED SHINY EXPERIENCE

Reset to Base Case

Flavor	Strawberry ▾	Holiday Special ▾
Price	$5.99 Supported range for 3 Scoops: $4.59 - $6.99	$5.79 Supported range for 1 Scoop: $3.99 - $6.99
Number of Scoops	3 Scoops ▾	1 Scoop ▾
Toppings	Sprinkles ▾	Caramel ▾

✓ Valid scenario — scenario results shown below

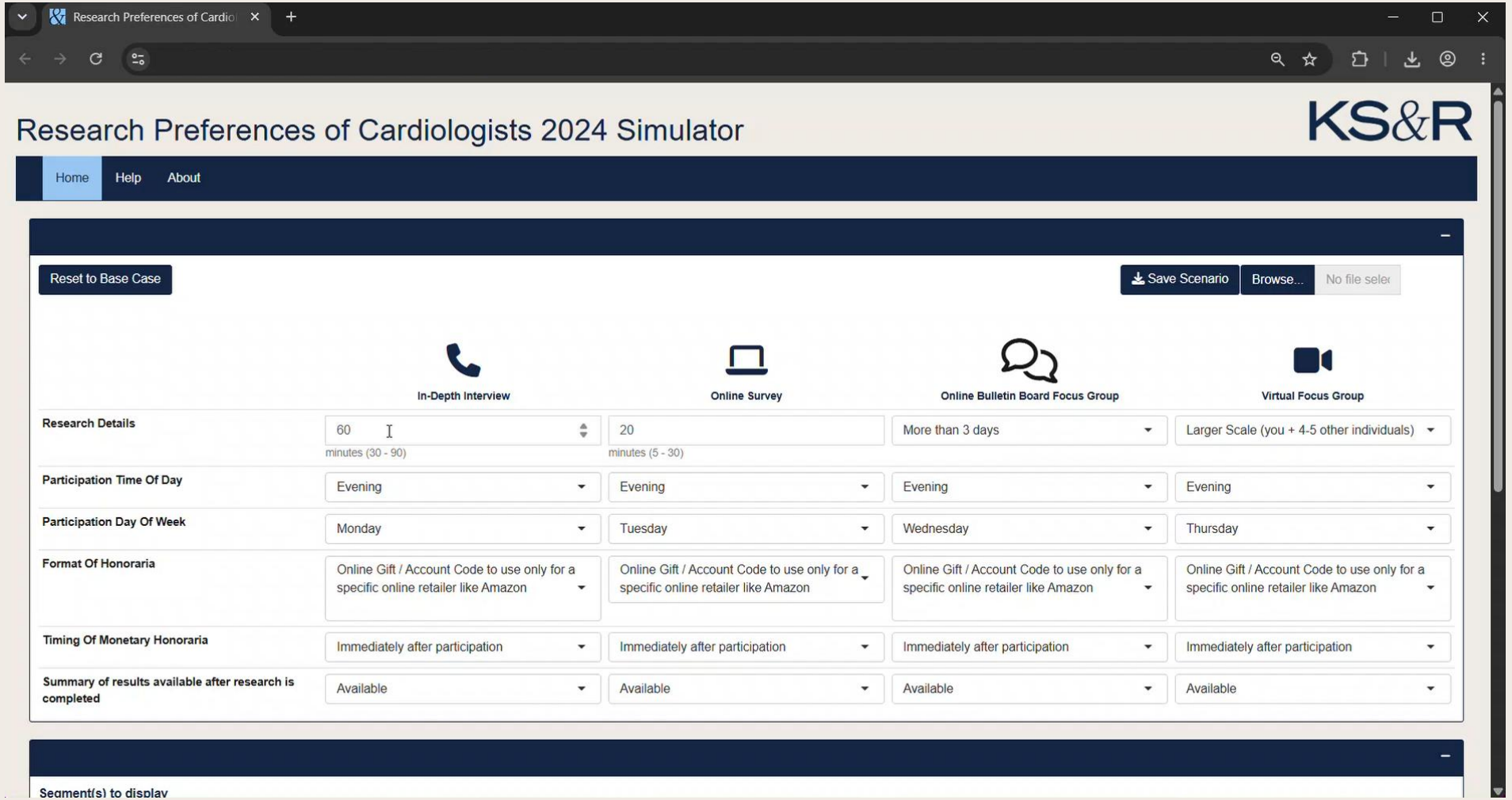
The underlying research model stays the same; the application adapts to the scenario being explored.

Demo: Example CBC Design Matrix

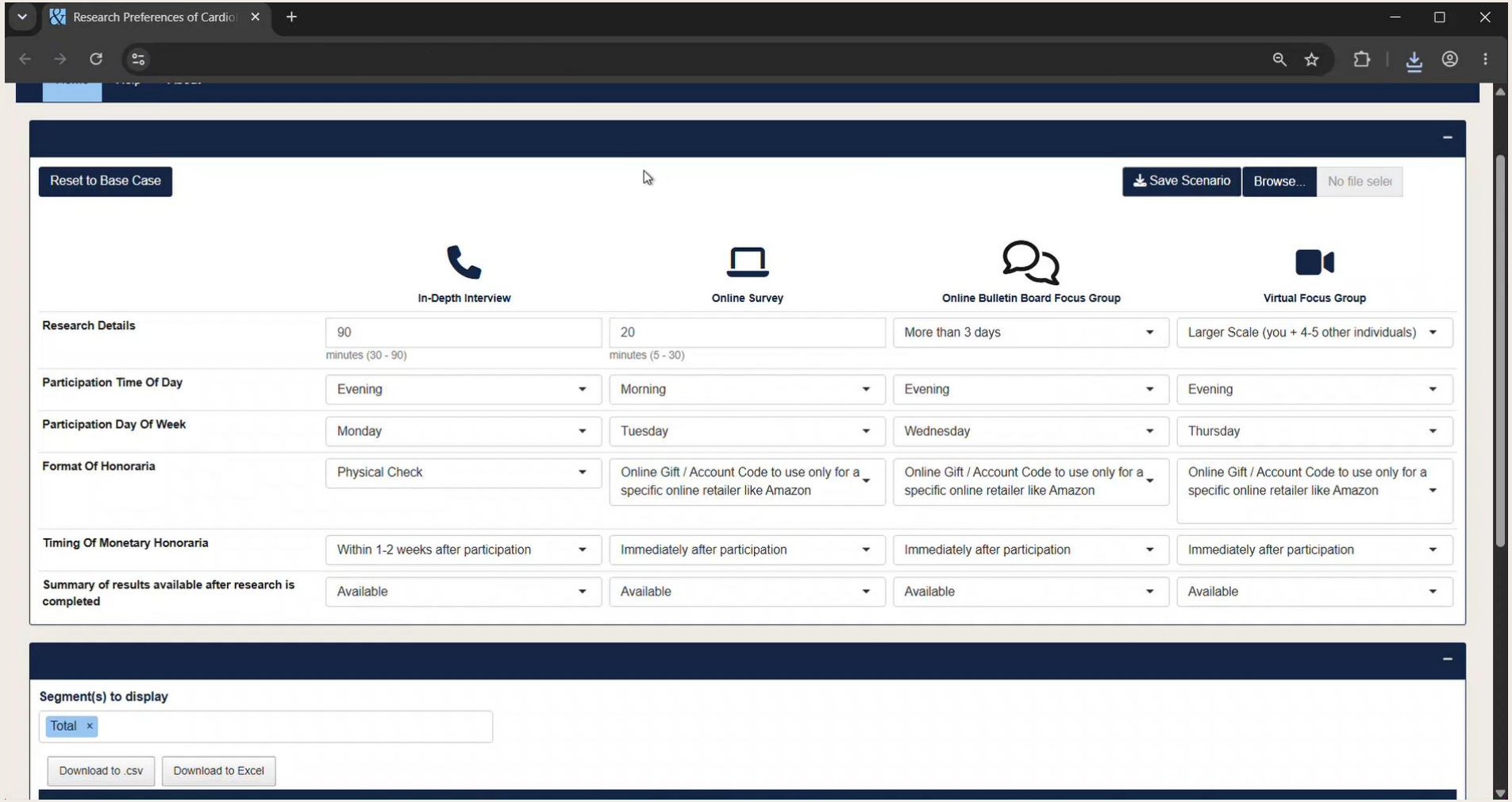
Cardiologists' preferences for participating in market research

Attribute	Applies To	Levels
Research Type	All	In-Depth Interview | Online Survey | Online Bulletin Board Focus Group | Virtual Focus Group
Time / length of In-Depth Interview	In-Depth Interview	30-minute | 45-minute | 60-minute | 90-minute
Time / length of Online Survey	Online Survey	5-minute | 10-minute | 20-minute | 30-minute
Duration of Online Bulletin Board Focus Group	Online Bulletin Board Focus Group	2 days | 3 days | More than 3 days
Size / # of people in Virtual Focus Group	Virtual Focus Group	Diad (you + 1 other individual) | Triad (you + 2 other individuals) | Larger Scale (you + 4-5 other individuals)
Participation Time Of Day	All	Morning | Afternoon | Evening
Participation Day Of Week	All	Monday | Tuesday | Wednesday | Thursday | Friday | Weekend
Format Of Honoraria	All	Physical Check | Physical Visa / Master Card Gift Card | Physical Gift Card from Online retailer like Amazon | Online Visa / Master Card Gift / Account Code to use for any online purchases | Online Gift / Account Code to use only for a specific online retailer like Amazon
Timing Of Monetary Honoraria	All	Immediately after participation | Within 24-48 hours after participation | Within 1-2 weeks after participation | Within 1 month after participation | Within 3 months after participation
Summary of results available after research is completed	All	Available | Not Available

Simulator Demo



Simulator Demo



Core Framework for a Shiny App

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The Shiny skeleton is surprisingly small

```r
library(shiny)
library(bslib)

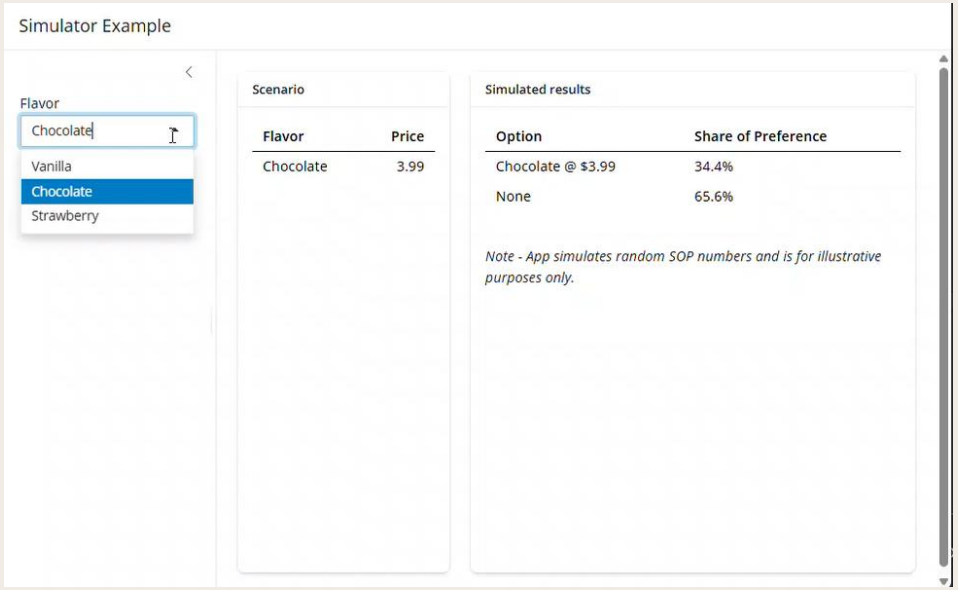
# Placeholder functions: replace with your own custom code and functions  ----------------------------
# Custom validation functions can be set up to perform error checking and return error messages
custom_validation_function <- function(scenario) {
  NULL
}
# Custom functions can be set to return a data frame of simulated results or other needed outputs
custom_calculation_function <- function(scenario) {
  # Placeholder: random share for the selected product; None takes the remainder
  product_share <- round(runif(1, min = 0.2, max = 0.9) * 100, 1)
  data.frame(Option = c(sprintf("%s @ $%.2f", scenario$Flavor, scenario$Price), "None"),
    `Share of Preference` = sprintf("%.1f%%", c(product_share, 100 - product_share)),
    check.names = FALSE
  )
}
# Can also pull in existing data to use in custom calculations, additional analyses, segment cuts, etc.
# utilities <- read.csv("./relevant_utility_data_file.csv") #commented out for this example code

# UI ----------------------------------------------------------------------
ui <- page_sidebar(
  title = "Simulator Example",
  sidebar = sidebar(
selectInput("flavor", "Flavor", choices = c("Vanilla", "Chocolate", "Strawberry")),
    numericInput("price", "Price", value = 3.99, min = 3.99, max = 5.99, step = 0.10)
  ),
  layout_columns(
    col_widths = c(4, 8),
    card(card_header("Scenario"), tableOutput("scenario")),
    card(card_header("Simulated results"), tableOutput("results"),
      tags$p(tags$em("Note - App simulates random SOP numbers and is for illustrative purposes only."))
    )
  )
)

# Server ------------------------------------------------------------------
server <- function(input, output, session) {
  scenario <- reactive({
    data.frame(Flavor = input$flavor, Price = input$price)
  })
  output$scenario <- renderTable({
    scenario()
  })
  output$results <- renderTable({
    msg <- custom_validation_function(scenario())
    validate(need(is.null(msg), msg))
    custom_calculation_function(scenario())
  })
}
shinyApp(ui, server)
```

A simple UI + server structure is enough to connect input controls to reactive behavior and display calculated outputs.

UI	What users see and select
SERVER	How the app responds
REACTIVE	When the experience updates



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WHAT THE CODE CONNECTS

1. **CSV INPUTS**
   Previously estimated utilities and study metadata
2. **DYNAMIC CONTROLS**
   Inputs change based on attribute levels selected
3. **CONSTRAINT CHECKS**
   Custom rules return helpful error messages
4. **CALCULATIONS → TABLE**
   Custom functions produce the displayed results

GitHub Repository

https://github.com/ksrinc/quant_ux_conf_2026_conjoint_simulators_in_shiny.git

A Good Interface Amplifies the Insights

Treat the simulator as a research product, not just an analytical output.

When all stakeholders can explore confidently, stay within what the data supports, and share what they find, the insight reaches further than the model alone could take it.

Thank you!



Connor Burton
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VP, DS&I

GitHub Repository

KS&R

KS&R

Dynamic Wisdom. Better Decisions.